JAN-P-466

28 APRIL 1947

JOINT ARMY-NAVY SPECIFICATION

POWDER, PHOTOFLASH

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army	Navy Department
50–11–102A 27 May 1943	.....

A. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWING

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–B–162—Barium Nitrate
JAN–B–217—Potassium Perchlorate
JAN–A–289—Aluminum Powder, Flaked, Grained, and Atomized (For Use in Ammunition)
JAN–M–382—Magnesium Powder (For Use in Ammunition)
JAN–M–454—Magnesium-Aluminum Alloy, Powdered

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition except Small Arms Ammunition[1]
100–2—Standard Specification for Marking Shipments by Contractors[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material[2]

FEDERAL SPECIFICATION

RR–S–366—Sieves; Standard, Testing

A–2. *Other publications.*—The following publications of the issue in effect on date of invitation for bids form a part of this specification.

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook[2]

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

A–3. *Drawing.*—The following drawing, of the issue in effect on date of invitation for bids, forms a part of this specification:

ARMY ORDNANCE DRAWING

76–4–43—Container, Black Powder, Storage and Shipping

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

707307—47—(259)

Source: https://assist.dla.mil -- Downloa
Check the source to verify that this is the

B. TYPES AND CLASSES

B–1. This specification covers the following types and classes of photoflash powder:

Type I—Mixed base of aluminum and magnesuim powders
Class A—Potassium perchlorate oxidizing agent
Type II—Magnesium-aluminum alloy powder base
Class A—Potassium perchlorate oxidizing agent
Class B—Barium nitrate oxidizing agent.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—The material used in the manufacture of photoflash powder shall conform to the requirements outlined in the following paragraphs:

C–1a. *Magnesium.*—Magnesium shall conform to the requirements for type I, grade A of Joint Army-Navy Specification JAN–M–382. There shall be no requirement with respect to apparent density and the granulation shall be as follows using U. S. Standard sieves conforming to Federal Specification RR–S–366:

Through No. 80 sieve	100 percent, minimum
Through No. 200 sieve	60 percent, minimum

C–1b. *Aluminum.*—Aluminum shall conform to the requirements for type A of Joint Army-Navy Specification JAN–A–289 except that the free metallic aluminum content shall be not less than 95.0 percent and the granulation shall be as follows:

Through No. 80 sieve	100 percent, minimum
Through No. 200 sieve	60 percent, minimum

C–1c. *Magnesium-aluminum alloy.*—Magnesium-aluminum alloy powder shall conform to the requirements for Type I of Joint Army-Navy Specification JAN–M–454 except that the granulation shall be as follows:

Through No. 80 sieve	100 percent, minimum
Through No. 200 sieve	60 percent, minimum

C–1d. *Potassium perchlorate.*—Potassium perchlorate shall conform to the requirements for grade A of Joint Army-Navy Specification JAN–P–217.

C–1e. *Barium nitrate.*—Barium nitrate shall conform to the requirements for class B of Joint Army-Navy Specification JAN–B–162 except that the granulation shall be as follows:

Through No. 100 sieve	100 percent, minimum
Through No. 200 sieve	95 percent, minimum

C–2. *Workmanship.*—The finished powder shall be a uniform mixture free from lumps or a tendency to form lumps.

D. GENERAL REQUIREMENTS

D–1. *Moisture.*—Maximum, 0.2 percent.

D–2. *Grease.*—Maximum, 1.0 percent.

D–3. *Granulation.*—

Through No. 80 sieve	100 percent, minimum
Through No. 200 sieve	60 percent, minimum

Source: https://assist.dla.mil -- Do
Check the source to verify that this is

E. DETAIL REQUIREMENTS

E–1. *Chemical composition.*—The chemical composition of photoflash powder shall conform to the requirements shown in table I.

TABLE I.—*Chemical composition*

	Type I	Type II	
	Class A	Class A	Class B
	Percent	*Percent*	*Percent*
Magnesuim	34.0 ± 1.0	.....	.....
Aluminum	26.0 ± 1.0	.....	.....
Magnesuim-aluminum alloy	.....	60.0 ± 1.0	44.5 ± 1.0
Potassium perchlorate	40.0 ± 1.0	40.0 ± 1.0	.....
Barium nitrate	.....	.....	55.5 ± 1.0

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lot.*—Maximum 25,000 pounds.

F–2. *Sampling.*—Select 10 percent but in no case more than 10 of the containers comprising the lot. By means of a scoop remove from each selected container a sufficient amount of material to form a primary sample of approximately 1 pound. Mix this sample thoroughly, place an 8-ounce portion in a tightly stoppered bottle, and label each primary sample so that the container from which it was taken can be easily identified. Mix the remaining portion of the primary samples thoroughly and quarter until a composite sample of about 1 pound is obtained. Place the composite sample in a tightly-stoppered bottle and label so as to show the type and class of the material, manufacturer, contract or order number and number of pounds in the lot. All acceptance tests shall be made on the composite sample representative of the lot. Hold the primary samples for possible future examination should the composite sample fail to meet the requirements.

F–3. *Inspection.*—

F–3a. *Army.*—Inspection shall be made in accordance with U. S. Army Specification 50–0–1 and shall be made at the point of delivery unless otherwise specified in the contract or order.

F–3b. *Navy.*—Inspection shall be made at the point of delivery unless otherwise specified in the contract or order.

F–4. *Tests.*—Tests to determine compliance with the requirements of this specification shall be made in accordance with the following paragraphs as applicable. For Navy purchases, the tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Moisture.*—Transfer a weighed portion of approximately 5 gm. of the sample to a tared weighing bottle. Dry the bottle and contents at 100° to 105° C. for 3 hours, cool in a desiccator, and weigh. Calculate the loss in weight as percent moisture.

F–4b. *Granulation.*—Place a weighed portion of approximately 10 gm. of the sample on the nest of sieves specified in paragraph D–3, properly superimposed and assembled with a cover and a bottom pan. Shake for 10 minutes by means of a mechanical shaker

Source: https://assist.dla.mil -- Downloa
Check the source to verify that this is the

which imparts to the sieves a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the material remaining on each sieve and calculate the percent passing through each sieve.

F–4c. *Grease.*—Transfer a weighed portion of approximately 10 gm. of the sample to a beaker, add 50 ml. of anhydrous ether, and stir. After a few minutes, filter the ether into a tared 250-ml. beaker. Wash the undissolved solids with two additional 50-ml. portions of anhydrous ether filtering each into the same beaker as the first. Evaporate the ether from the beaker by heating or by a current of dry air. Heat the beaker and residue at 60° C. for an hour, cool in a desiccator, and weigh. Calculate the increase in weight as percent grease on a moisture-free basis.

F–4d. *Composition.*—

F–4d(1). *Type I, class A.*—

F–4d(1)a. *Potassium perchlorate.*—Transfer a weighed portion of approximately 2 gm. of the sample to a 250-ml. beaker and surround this with ice water. Add 100 ml. of distilled water at a temperature of 0° to 5° C. Stir the mixture vigorously for 8 minutes scraping down the inner wall of the beaker. Transfer the residue to a tared Gooch crucible and wash thoroughly with distilled water at 0° to 5° C. Wash the residue with alcohol, then with ether, dry the crucible and contents at 100° to 105°C. for 1 hour, cool in a desiccator, and weigh. Calculate the loss in weight of the sample as percent potassium perchlorate on a moisture-free basis.

F–4d(1)b. *Magnesium.*—Extract the residue from the potassium perchlorate determination with 5-percent acetic acid until all magnesium is dissolved, as indicated by no evolution of gas when the residue is treated with fresh portions of the acetic acid solution. Wash the undissolved residue with distilled water, rinse with alcohol and then with ether, heat the crucible and contents for an hour at 60° C., cool in a desiccator, and weigh. Calculate the loss in weight as percent magnesium on a moisture-free basis.

F–4d(1)c. *Aluminum.*—Calculate the weight of undissolved residue in the crucible (see par. F–4d(1)b) as percent aluminum on a moisture-free basis.

F–4d(2). *Type II, class A.*—

F–4d(2)a. *Potassium perchlorate.*—See paragraph F–4d(1)a.

F–4d(2)b. *Magnesium-aluminum alloy.*—Calculate the weight of the undissolved residue in the crucible remaining from the determination of the potassium perchlorate as percent magnesium-aluminum alloy on a moisture-free basis.

F–4d(3). *Type II, class B.*—

F–4d(3)a. *Barium nitrate.*—Use the same procedure as specified for potassium perchlorate in paragraph F–4d(1)a.

F–4d(3)b. *Magnesium-aluminum alloy.*—See paragraph F–4d(2)b.

F–5. *Resubmission.*—If the lot or any portion of the lot fails to pass the inspection tests, the manufacturer shall have the option

Source: https://assist.dla.mil -- Do
Check the source to verify that this is

of having additional analyses made, without expense to the Government, to effect complete segregation of all defective portions of the lot. The manufacturer may then remove or replace defective portions of the lot and resubmit the lot for acceptance, provided that the markings on the containers are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the Government inspector. New samples shall be taken from the entire resubmitted lot and subjected to all of the inspection tests required by this specification.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging and packing.*—Unless otherwise specified, packaging and packing shall conform to the requirements for shipment of inflammable solids and oxidizing materials as specified in Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc. In addition, packing containers shall conform to Army Ordnance Drawing 76–4–43 or a commercial container of equal quality. A paper bag liner shall be used. A gasket which completely covers the opening shall be provided to prevent any possibility of loose powder becoming lodged in the screw heads. The container shall be painted black.

G–2. *Marking.*—In addition to any special marking required by the contract or order and marking to insure safe handling as required by the Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., shipments for the Army shall be marked in accordance with the requirements of the U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—Photoflash powder is intended for use in photoflash ammunition, miscellaneous Army photographic uses, and for special effects.

H–2. Requests, requisitions, schedules, and contracts or orders should contain the following:

(a) Title, number, and date of the specification.
(b) Type and class required. (See par. B–1.)

H–3. Copies of Army Ordnance Drawing 76–6–43 may be obtained from Office, Chief of Ordnance, War Department, Washington 25, D. C.

H–4. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

Source: https://assist.dla.mil -- Downloa
Check the source to verify that this is the

H–5. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–6. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, War Department, Washington 25, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Army: OES.
Navy: OS.

[JAN-P-466]

Source: https://assist.dla.mil -- Do
Check the source to verify that this is